
STOR 320: Final Project Presentation



Question

How do demographic and study-related factors predict students' GPA, and do these relationships differ between the USA and Australia?

Why is this interesting?

- Differences in cultural, social, and institutional conditions contribute to unique academic experiences
- Answering this question:
 - Provides insight into how universal academic success factors are
 - Highlights cross-cultural differences between distinct educational systems
 - Connects individual student behavior to broader educational trends

Data Source

U.S. Student Data

- **2392** American high school students
- Pulled Age, Gender, GPA, Parental Income, and Weekly Study Hours columns
- Originally compiled and published on Kaggle by Rabie Kharoura in 2024 (CC4.0 International)

Australian Student Data

- 10000+ Australian university students
- Selected only students aged 18 and under (**6653** students)
 - Randomly sampled **2392**
- Pulled Age, Gender, High School GPA, Parental Income, and Weekly Study Hours columns
- Originally compiled and published on Kaggle by Laoshi in 2024 (CC0 Public Domain)

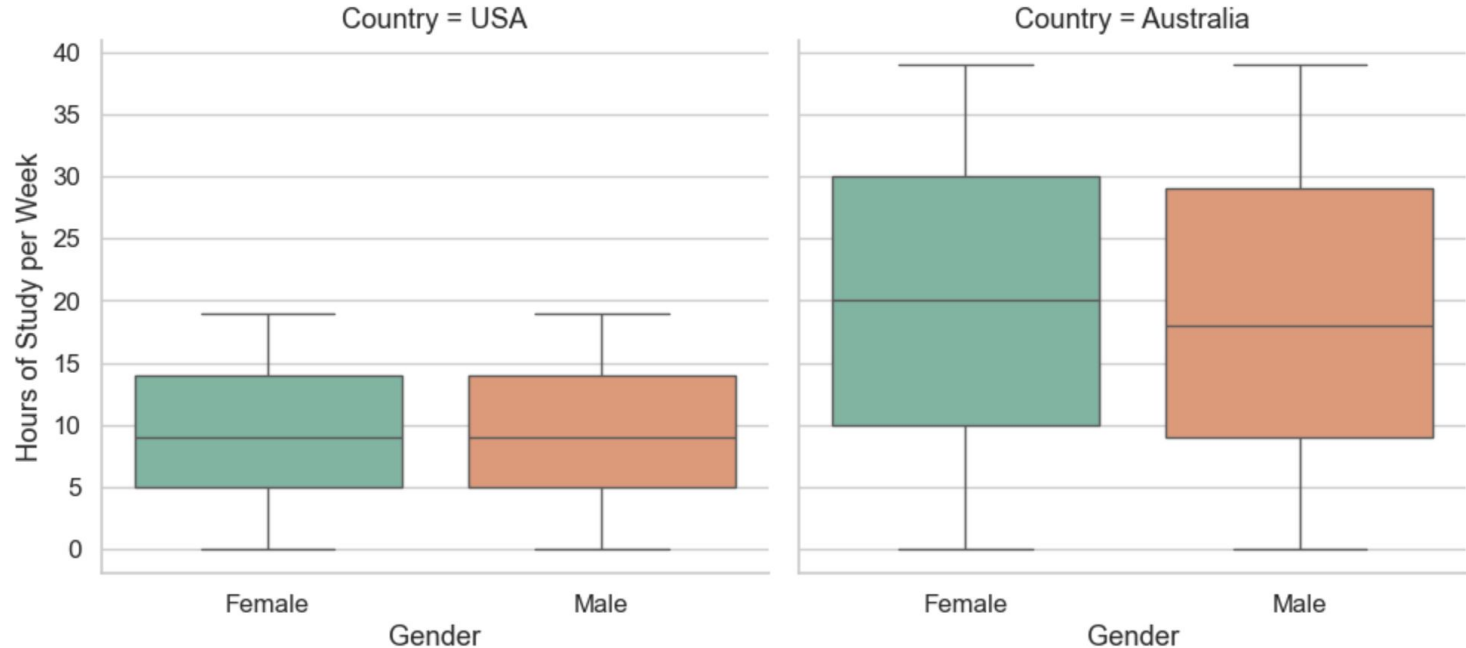
Descriptive statistics

US & Australian Student dataset

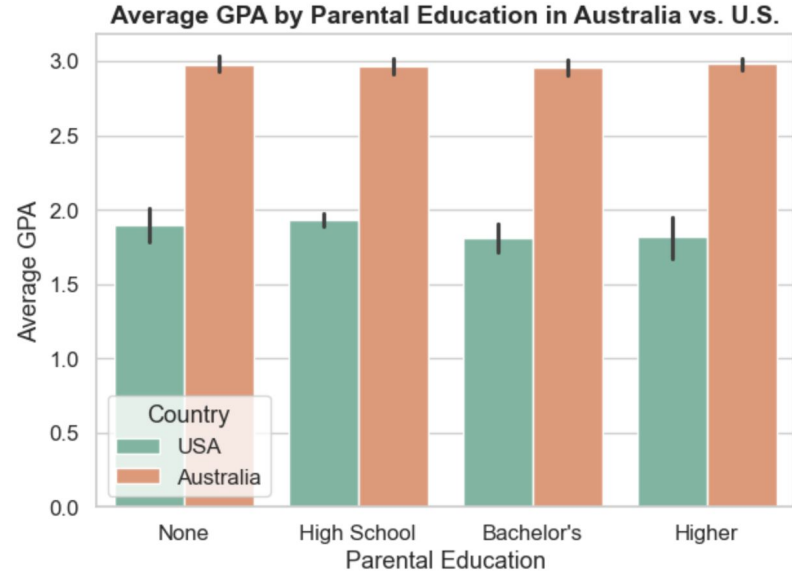
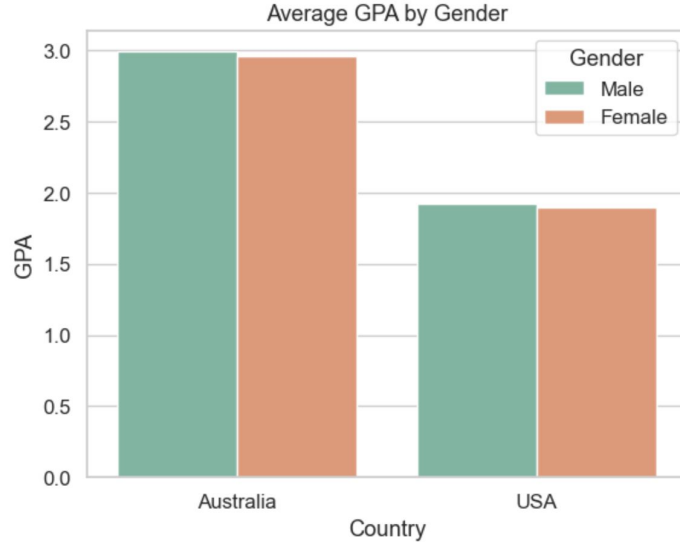
- 2392 American students and 2392 Australian students
- Age range: 15–18 years old
- Australian dataset:
 - 1,178 female and 1,214 male students
 - Mean GPA: 2.97 ($s = 0.59$)
- American dataset:
 - 1,170 female and 1,222 male students
 - Mean GPA: 1.9 ($s = 0.92$)

Descriptive statistics

Study Hours per Week by Gender and Country



Descriptive statistics



Method 1: Multiple Linear Regression

- Estimated two multiple linear regression models using:
 - **Model 1: $GPA \sim StudyHours * Country$**
 - Tests whether study hours predict GPA and whether the slope differs between USA and Australia (interaction term)
 - **Model 2 (with controls): $GPA \sim StudyHours * Country + Age + Gender + ParentalEducation$**
 - Added demographic variables Age, Gender, Parental Education
 - Evaluated whether demographics improve prediction accuracy or help explain GPA differences across countries
- All predictors entered as **fixed effects** (each variable in the regression model is assumed to have a single, constant effect on GPA across all students in the dataset)

Method 2: Ordinary Least Squares (OLS)

- **OLS estimation provided:**
 - Unbiased coefficient estimates (β 's)
 - Standard errors
 - t-statistics and p-values for hypothesis testing
- Tested for country differences in the StudyHours slope using the **p-value of the interaction term**
- Used **t-tests** to evaluate significance of individual predictors
- Used the model's **F-statistic** to assess overall fit
- Checked whether interaction terms and demographic variables produced significant **improvements in R^2**
- OLS framework enabled a **clear comparison of GPA predictors** across national contexts

Results pt. 1

- **Model Fit**
 - Regression explained **35% of GPA variation** ($R^2 = 0.35$), indicating a moderate fit.
- **Country Differences**
 - **Country was the strongest predictor** of GPA.
 - U.S. students scored **1.32 GPA points lower** than Australian students on average ($\beta = -1.3203$, $p < .001$).
- **Study Hours**
 - Study Hours alone were not a significant predictor across all students ($\beta = 0.0015$, $p = .255$).

Results pt. 2

- **Different Effects by Country**
 - **U.S. students:** Each additional study hour predicted a **0.027 GPA increase**.
 - **Australian students:** No meaningful relationship between study hours and GPA.
- **Demographic Predictors**
 - Age, Gender, and Parental Education were not significant ($p > .17$).
 - These demographics do not meaningfully explain GPA differences across countries.
- **Overall Conclusion**
 - Study behaviors matter for GPA in the U.S. but not in Australia.
 - Country differences are far more influential than study hours or demographics in predicting GPA.

Results pt. 3



- Australian students consistently earn higher GPAs across all study-hour levels, while U.S. students show lower GPAs even at similar study hours.
- The separation between the two groups supports the idea that country is a stronger predictor of GPA than study hours.



Thanks!

Do you have any questions?